

CS & IT ENGINEERING



Digital Logic

Miscellaneous Topics

Lecture No. 1



By- Aditya sir

Recap of Previous Lecture



Topic



Synchronous Counters



Topics to be Covered



Topic

Number System





About Aditya Jain sir



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.





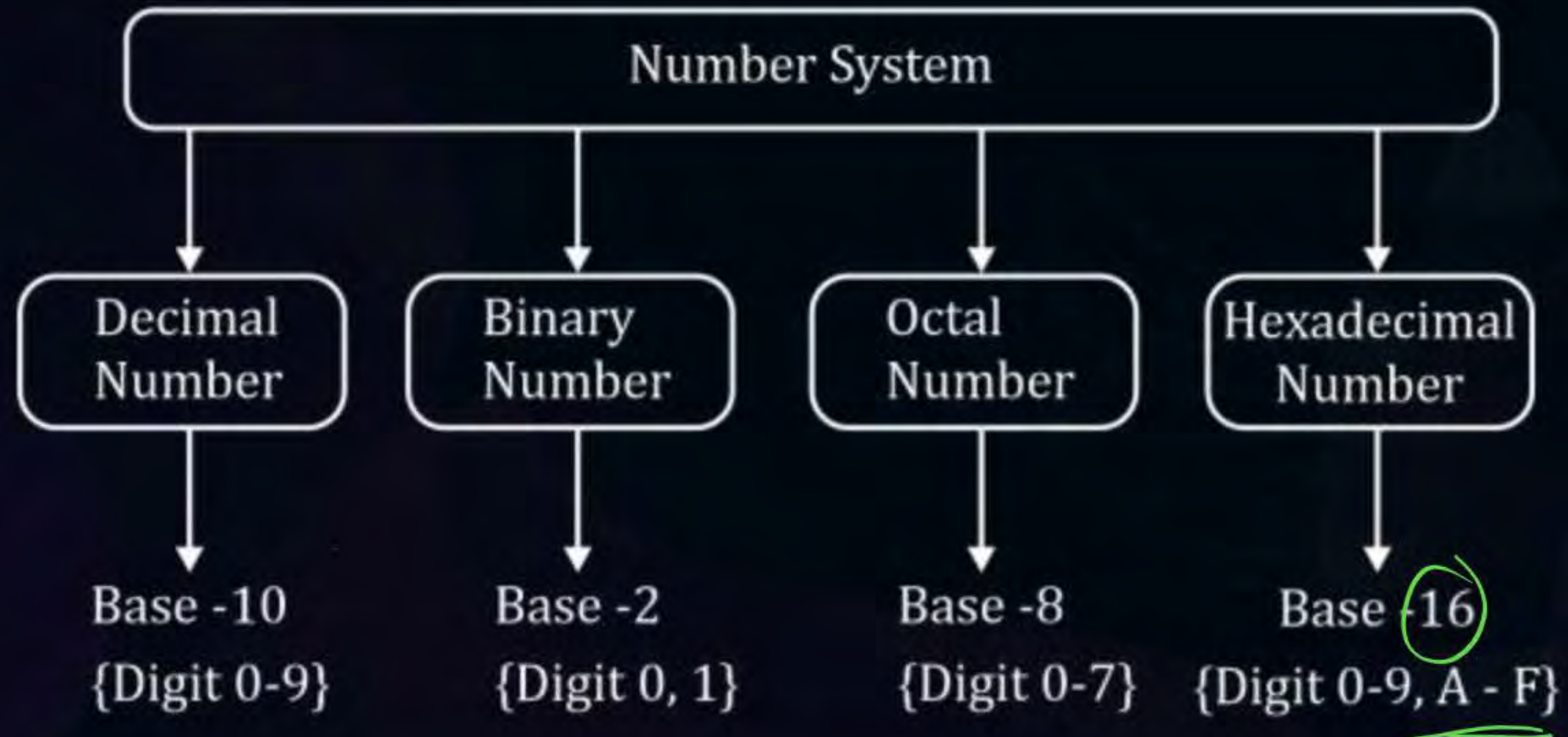
Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic : Number System

Base (Radix)

Total number of digit used in the system

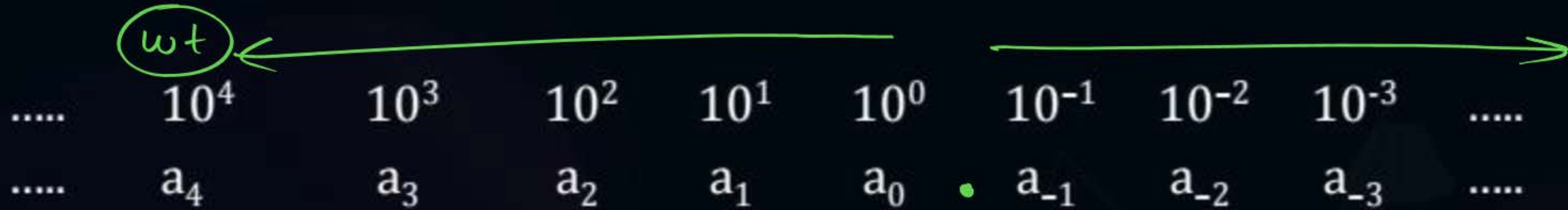




Topic : Number System

Base (Radix)

Decimal:



$a \rightarrow$ Coefficient of decimal number system

$10 \rightarrow$ Weight of decimal number system

$\downarrow$
Importance



Topic : Number System

Example:

$$(501.23)_{10}$$

10^2	10^1	10^0	10^{-1}	10^{-2}
5	0	1	2	3

Weight $\rightarrow 10^2 \quad 10^1 \quad 10^0 \quad 10^{-1} \quad 10^{-2}$

Coefficient $\rightarrow (\underline{501.23})_{10}$

5 0 1 . 2 3



Topic : Number System

Base	Digit	V. imp
→ 2	0, 1	
3	0, 1, 2	
4	0, 1, 2, 3	
5	0, 1, 2, 3, 4	A → 10
6	0, 1, 2, 3, 4, 5	B → 11
7	0, 1, 2, 3, 4, 5, 6	C → 12
8	0, 1, 2, 3, 4, 5, 6, 7	D → 13
9	0, 1, 2, 3, 4, 5, 6, 7, 8	E → 14
10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9	F → 15
11	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A	
12.	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B	
13.	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C	
14.	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D	
15.	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E	
16.	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F	

$$0 \rightarrow b-1$$

(101)
2, 3, 4, ...



Topic : Number System

Binary Number System (Base (Radix) = 2)

.....	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}	
.....	a_4	a_3	a_2	a_1	a_0	a_{-1}	a_{-2}	a_{-3}	

$2^i \rightarrow$ Weight of Binary number system

$a_i \rightarrow$ Coefficient of Binary number system {0,1}

$0 \rightarrow b-1$



Topic : Number System

Example:

$(101.11)_2$ *Binary*

2^2	2^1	2^0	2^{-1}	2^{-2}
1	0	1	1	1



Topic : Number System

Octal Number System (Base (Radix) = 8)

.....	8^4	8^3	8^2	8^1	8^0	8^{-1}	8^{-2}	8^{-3}	
.....	a_4	a_3	a_2	a_1	a_0	a_{-1}	a_{-2}	a_{-3}	

$8^i \rightarrow$ Weight of Octal Binary number system

$a_i \rightarrow$ Coefficient of Binary number system $\{0,7\}$



Topic : Number System



Example:

$$(720.64)_8$$

8^4	8^1	8^0	8^{-1}	8^{-2}
7	2	0	6	4



Topic : Number System

Hexadecimal Number System (Base (Radix) = 16)

$$\begin{array}{ccccccccccccc} & \longleftarrow & & & & & & & & & & & \longrightarrow \\ \text{.....} & 16^4 & 16^3 & 16^2 & 16^1 & 16^0 & 16^{-1} & 16^{-2} & 16^{-3} & \text{.....} \\ \text{.....} & a_4 & a_3 & a_2 & a_1 & a_0 & a_{-1} & a_{-2} & a_{-3} & \text{.....} \end{array}$$

Hexadecimal

$16^i \rightarrow$ Weight of ~~Binary~~ number system

$a_i \rightarrow$ Coefficient of Binary number system {0 - 9, A - F}

10-15
==

$(A2C.F)_{16}$



Topic : Number System

Example:

$$(A2C.F)_{16}$$

16^2	16^1	16^0	16^{-1}
A	2	C	F



In base conversion 2 key points are there:

- } Conversion

(Q.2) $(n)_{10} \longrightarrow (?)_6$

(0.3) $(n)_{b_1} \rightarrow ()_{b_2}$

$\searrow \quad \nearrow$

$()_{10}$



Topic : Number System

In base conversion 2 key points are there:

$$(\quad)_r \longrightarrow (\quad)_{10}$$

(A) Any base to Decimal conversion

$$(a_3 \overset{r^3}{\quad} a_2 \overset{r^2}{\quad} a_1 \overset{r^1}{\quad} a_0 \overset{r^0}{\quad} . \quad a_{-1} \overset{r^{-1}}{\quad} a_{-2} \overset{r^{-2}}{\quad}) = (\quad)_{10}$$

$$(a_3 \times r^3 + a_2 \times r^2 + a_1 \times r^1 + a_0 \times r^0 + a_{-1} \times r^{-1} + a_{-2} \times r^{-2}) = (\quad)_{10}$$



Topic : Number System

Case-1: Binary to Decimal conversion

Example: $(1011.11)_2 = (\quad)_{10}$

Handwritten powers of 2 above the binary digits:
3 2 1 0 -1 -2
2 2 2 2 2 2

$$\Rightarrow [(1 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) + (1 \times 2^{-1}) + (1 \times 2^{-2})]_{10}$$

$$[8 + 0 + 2 + 1 + 0.5 + 0.25]_{10}$$

$$\Rightarrow (11.75)_{10}$$

010
Y

$$(729.4)_8 \rightarrow (?)_{10}$$



invalid

$$(721.4)_8 \rightarrow (?)_{10}$$



Topic : Number System

Case-2: Octal to Decimal conversion

Example: $(721.4)_8 = (\quad)_{10}$

$\begin{matrix} 8^2 & 8^1 & 8^0 & 8^{-1} \\ 7 & 2 & 1 & 4 \end{matrix}$

$$\Rightarrow [(7 \times 8^2) + (2 \times 8^1) + (1 \times 8^0) + (4 \times 8^{-1})]_{10}$$

$$\Rightarrow [448 + 16 + 1 + 0.5]_{10}$$

$$\Rightarrow \underline{\underline{(465.5)_{10}}}$$

$$(A2B.C)_{16} \rightarrow (?)_{10}$$



Topic : Number System

Case-3: Hexadecimal to Decimal conversion

Example: $(A2B.C)_{16} = ()_{10}$

Handwritten green annotations above the example:
2, 1, 0, -1 above the digits A, 2, B, and C respectively.
16, 16, 16, 16 below the exponents 2, 1, 0, -1 respectively.

$$\Rightarrow [(A \times 16^2) + (2 \times 16^1) + (B \times 16^0) + (C \times 16^{-1})]_{10}$$

$$\Rightarrow [(10 \times 256) + (2 \times 16) + (11 \times 1) + (12 \times 16^{-1})]_{10}$$

$$\Rightarrow [2560 + 32 + 11 + 0.75]_{10}$$

$$\Rightarrow (2603.75)_{10}$$

Handwritten green box:
 $A \rightarrow 10$



Topic : Number System

Case-4: Base 5 to Decimal conversion

Example: $(432.22)_5 = ()_{10}$

$$\Rightarrow [(4 \times 5^2) + (3 \times 5^1) + (2 \times 5^0) + (2 \times 5^{-1}) + (2 \times 5^{-2}) +]_{10}$$

$$\Rightarrow [100 + 15 + 2 + 0.4 + 0.08]_{10}$$

$$\Rightarrow (117.48)_{10}$$

$$(4365.23)_7 \longrightarrow (?)_{10}$$

[NAT]



#Q. $(4365.23)_7 = (?)_{10}$

$$\Rightarrow [(4 \times 7^3) + (3 \times 7^2) + (6 \times 7^1) + (5 \times 7^0) + (2 \times \underline{7^{-1}}) + (3 \times \underline{7^{-2}})]_{10}$$

$$\Rightarrow \underline{\underline{(1566.346)_{10}}}$$

[NAT]



Tricky

#Q. $(62C) = (?)_{10}$

By minimum base convert into decimal?

37.5%

$(62C)$ ✓
13, 14, 15, ...

A $\rightarrow 10$

B $\rightarrow 11$

$C \rightarrow 12$

base = (12)

6 2 C
 13^2 13^1 13^0

$$6 \times 13^2 + 2 \times 13^1 + 12 \times 1 = (1052)_{10}$$

②

Decimal $\longrightarrow$ Any base

$$(n)_{10} \longrightarrow (?)_b$$

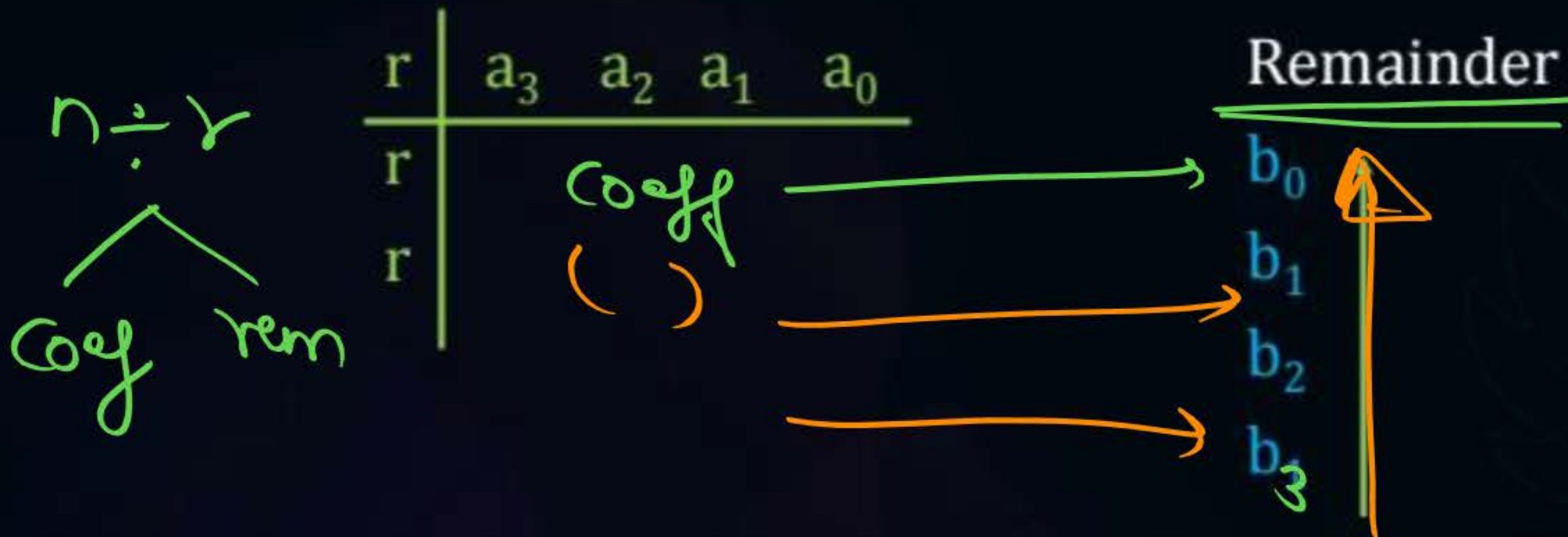


Topic : Number System

In base conversion 2 key points are there:

(B) Decimal to any other base conversion

$$(n)_{10} \longrightarrow (?)_r$$



$$(a_3 a_2 a_1 a_0 a_{-1} a_{-2} a_{-3})_{10}$$



Topic : Number System

0.75

$$0. a_1 a_2 a_3 \times r = X_0 \cdot X_1 X_2$$

$$0. x_1 x_2 \times r = X_1 \cdot X_3 X_4$$

$$0. x_3 x_4 \times r = X_2 \cdot X_5 X_6$$

$$(a_3 a_2 a_1 a_0 \cdot a_1 a_2 a_3)_{10} = (b_3 b_2 b_1 \cdot x_0 x_1 x_2)_r$$

Final

Q9

$$(723.25)_{10} \rightarrow ()_3$$



$$0.25 \times 3 = 0.75$$

$$0.75 \times 3 = 2.25$$

$$0.25 \times 3$$

Ans
0.2

① eg:-

$$(19.75)_{10} \longrightarrow (?)_2$$

Telegram



Topic : Number System

Case-1: Decimal to Binary Base conversion.

Example: $(19.75)_{10} = ()_2$

Before Decimal

After Decimal

2	19	<u>Rem</u> 1
2	9	1
2	4	0
2	2	0
	1	1
	0	Stop

$$\begin{aligned} 0.75 \times 2 &= 1.5 && 1 \\ 0.5 \times 2 &= 1.0 && 1 \end{aligned}$$

$(10011.11)_2$

$$(19.75)_{10} = (10011.11)_2$$

(eg2)

$$(210.23)_{10} \longrightarrow (?)_8$$



Topic : Number System

$$\div 8 \Rightarrow \overset{\text{rem}}{\textcircled{0-7}}$$

Case-2: Decimal to Octal Base conversion.

Example: $(\overset{210.23}{\cancel{210.23}})_{10} = (\quad)_8$

Before Decimal
Remainder

After Decimal

8	210	2
8	26	2
	3	3
	0	

$$\begin{aligned} 0.23 \times 8 &= 1.84 & 1 \\ 0.84 \times 8 &= 6.72 & 6 \\ 0.72 \times 8 &= 5.72 & 5 \\ 0.72 \times 8 & & \end{aligned}$$

$(322.165)_8$

$$(210.23)_{10} = (322.165)_8$$

Challenge

$$\text{eg3)} \quad (1228.55)_{10} \longrightarrow (?)_{16}$$



Topic : Number System

A $\rightarrow$ 10
B $\rightarrow$ 11
C $\rightarrow$ 12

Case-3: Decimal to Hexadecimal Base conversion.

Example: $(1228.55)_{10} = ()_{16}$

Before Decimal
Remainder

After Decimal

16	1228	12(C)
16	76	12(C)
	4	4
	0	

$$0.55 \times 16 = 8.84 \quad 8$$
$$0.84 \times 16 = 12.8 \quad 12(C)$$

$(4CC.8C)_{16}$

$$(1228.56)_{10} = (4CC.8C)_{16}$$

[NAT]



#Q. $(93.25)_{10} = (?)_5$

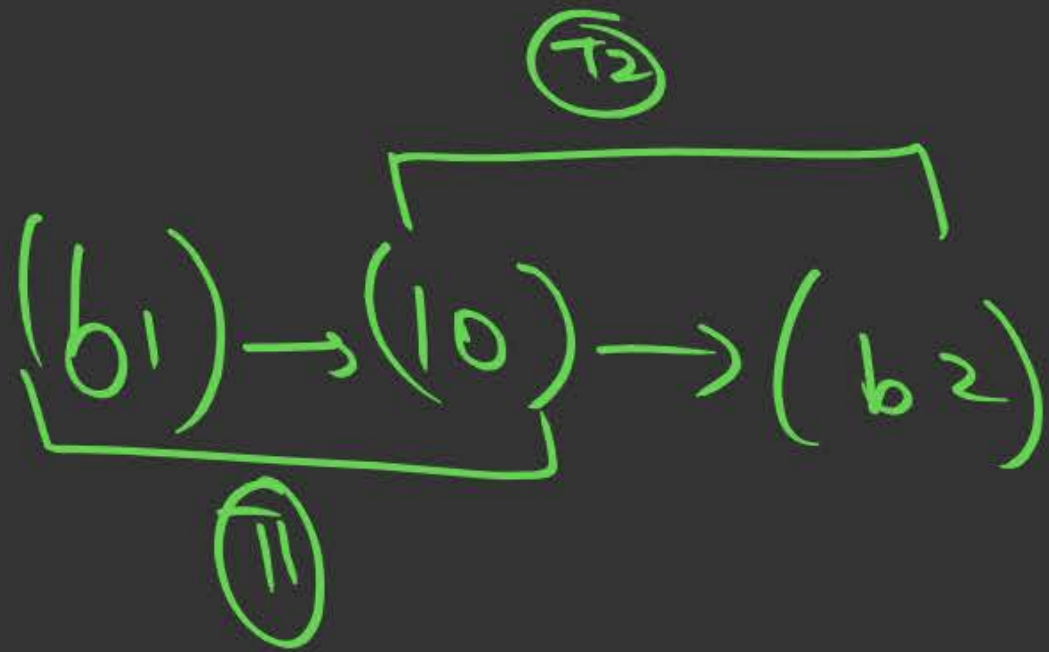
	Rem
5 93	3
5 18	3
5 3	3
5 0	Stop

$(333.11)_5$

$$0.25 \times 5 = 1.25$$

$$0.25 \times 5 = 1.25$$

Type 3 :- Any base $\longrightarrow$ Any other Base





Topic : Number System

$$\underline{\text{How}} \quad ()_2 \rightarrow ()_{10} \rightarrow ()_8$$

Some Special Case $\Rightarrow$

only for

$$2^x \longleftrightarrow 2^y$$

Case-1: Binary to Octal base conversion

Example: $(10110111)_2 = ()_8$

Octal $\rightarrow$ means base 8

$$8 = 2^3 \checkmark$$

$$()_2 \rightarrow ()_{10} \rightarrow ()_8$$

$$2^1 \rightarrow 2^3 \checkmark$$

Every three digits of binary represent one digit of octal

(Right to left)

$$\begin{array}{ccc} (010) & (110) & (111) \\ \downarrow & \downarrow & \downarrow \\ 2 & 6 & 7 \end{array}$$

$$\text{Hence, } (10110111)_2 = ()_8 \Rightarrow (267)_8$$



Topic : Number System

HW

$$(\)_2 \rightarrow (\)_{10} \rightarrow (\)_{16}$$

$$2^1 \rightarrow 2^4 \checkmark$$

Some Special Case

Case-2: Binary to Hexadecimal base conversion

Example: $(1011011)_2 = (\)_{16}$

Hexadecimal $\rightarrow$ means base 16

$$16 = 2^4$$

Every four digits of binary represent one digit of Hexadecimal.

0101 1011

5

11(B)

Hence, $(1011011)_2 = (\)_{16}$

$(0101) \rightarrow 5$

$(1011) \rightarrow 11 \rightarrow B$

$\Rightarrow (5B)_{16}$

[NAT]



#Q. $(3221)_4 = (\)_2$

App) $(\)_4 \rightarrow (\)_{10} \rightarrow (\)_2 \Rightarrow \text{HW}$

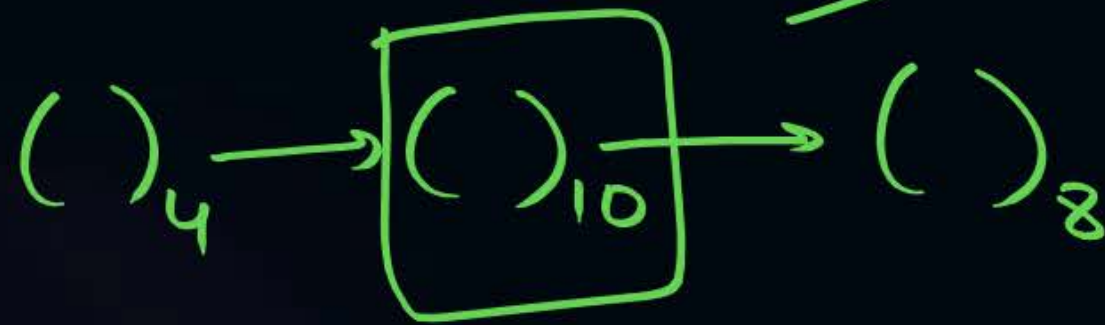
Shortcut

$$4 = 2^2$$

$$\begin{array}{cccc} (3 & 2 & 2 & 1)_4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ (11 & 10 & 10 & 01)_2 \Rightarrow \end{array}$$
$$\begin{array}{cccc} & & & (3 & 2 & 2 & 1)_4 \\ & & & \uparrow & & & \\ & & & (11 & 10 & 10 & 01)_2 \\ & & & \underline{\underline{\hspace{1cm}}} & & & \end{array}$$

#Q. $(330123)_4 = ()_8$

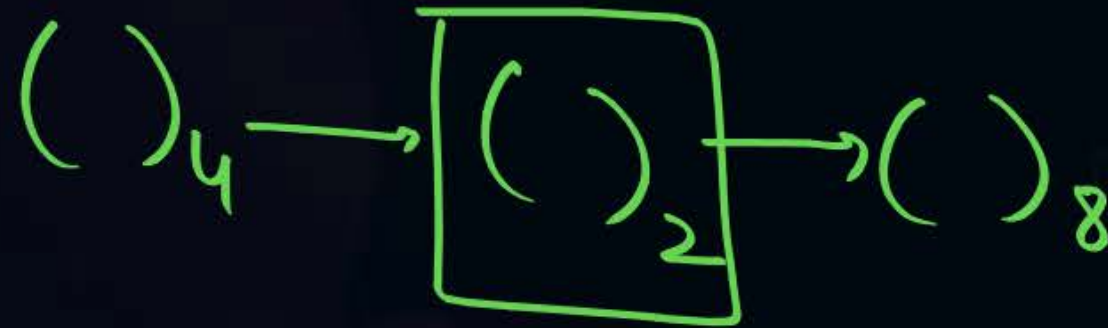
Appro 1:



universal

(Hw) ✓

Appro 2:



$$(330123)_4 \rightarrow (?)_2$$

$$4 = 2^{\textcircled{2}}$$

>

$$\begin{array}{cccccc} 3 & 3 & 0 & 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ (11 & 11 & 00 & 01 & 10 & 11)_2 \rightarrow (?)_8 \\ \downarrow & \downarrow & \downarrow & \downarrow & & \\ 7 & 4 & 3 & 3 & & \end{array}$$

$$2 \rightarrow 2^{\textcircled{3}}$$

$$\Rightarrow (7433)_8 \checkmark$$

[NAT]



#Q. $(7634)_8 = (\)_{16}$

$$(\)_8 \rightarrow (\)_2 \rightarrow (\)_{16}$$

$$16 = 2^4$$

① (7634)
 $(\begin{smallmatrix} 111 & 110 & 011 & 100 \end{smallmatrix})_2 \Rightarrow$

$(\begin{smallmatrix} 111 & 110 & 011 & 100 \end{smallmatrix})_2 \rightarrow (\)_{16}$
 $\downarrow \quad \downarrow \quad \downarrow$
 $(1111) \quad (1001) \quad (1100)$
 $\downarrow \quad \downarrow \quad \downarrow$
 $F \quad 9 \quad C$
 $\Rightarrow \underline{\underline{(F9C)_{16}}}$

[MCQ]



#Q. Find number of solutions of x and y which can satisfy the equation $(43)_8 = (x8)_y$

HW

A

1

B

2

C

3

D

4

[MCQ]



#Q. The result of addition operation $34 + 43$ performed on minimum base is stored in an 8-bit register. The content of register will be

- A** 01000011
- B** 00101010
- C** 01010101
- D** 01010100

$$\begin{matrix} (34) & + & (43) \\ 5 & & 5 \end{matrix}$$

HW

90%

010 50%



2 mins Summary



$$\rightarrow ()_b \rightarrow ()_{10}$$

$$\rightarrow ()_{10} \rightarrow ()_b$$

$$\rightarrow ()_{b_1} \rightarrow ()_{b_2}$$



THANK - YOU